🛰️ AEF Crop Intelligence: User Guide

AEF Crop Intelligence is a decision support tool for farmers and agronomists. It combines a **Digital Twin** of your field (a mechanistic computer simulation of crop growth) with **Satellite Intelligence** (real-time observations from space) to help you optimize yield, manage diseases, and reduce input costs.

It answers three critical questions:

1. **How is my crop doing** compared to its biological potential?
2. **Where exactly** are the disease/stress hotspots?
3. **What specific actions** (irrigation, fertilization, pruning) should I take?

🔐 **Privacy Note:** The Access Code required at login is **not** an account creation. It is simply a key to unlock the tool for authorized testers. We do not save your farm data, field coordinates, or production history on our servers. You have the option to save your own configuration locally (`.json`), but your data remains yours.

🚀 Getting Started

The app guides you through a 5-step configuration process to build your field's Digital Twin.

Step 1: 🌍 Geography

Define exactly where your field is located. The new **Assisted Setup** makes this easier than ever.

- **Tab 1: ✨ Assisted Setup (Recommended)**
  - **Input:** Enter the exact coordinates (e.g., `4°34′ N 11°07′ E` or `4.56, 11.12`) and the estimated **Field Area** in hectares.
  - **Smart Scan:** Click **Generate Smart Field**. The system will scan the entered location *and* surrounding areas to automatically find a valid plot of land that avoids buildings, roads, and water bodies.
  - **Validation Status:**
    - 🟢 **SAFE:** Valid vegetation/soil detected.
    - 🟡 **WARNING:** Some non-crop features detected (e.g., a path). You can proceed but should double-check boundaries.
    - 🔴 **CRITICAL:** High density of buildings or deep water. You should likely move the field.
  - **Fine-Tune:** Use the **Nudge Buttons** (⬅️ ⬆️ ⬇️ ➡️) to shift the generated box precisely over your field on the map.
- **Tab 2: ✍️ Manual Draw**
  - Use this tab if the auto-generated shape isn't perfect. Select the **Polygon Tool** (pentagon icon) on the map to manually trace your field boundaries vertex-by-vertex. The Satellite view helps you see fences and treelines.

- **Tab 3: 📂 Load Config**
    - Upload a previously saved `field_config.json` file to restore your settings instantly.

Step 2: 🌱 Crop System

Tell the system what you are growing.

- **Select Crop:** Choose from major Annuals (Maize, Cotton, Rice, Wheat, Soybean, Cassava) or Perennials (Cocoa, Coffee).
- **Variety:** Select the specific variety (e.g., "Hybrid" vs "Landrace"). This updates the genetic parameters of the simulation.
- **Planting Date:**
    - **For New Fields:** Set the date you plan to plant.
    - **For Established Fields:** Set the *original* planting date. The Digital Twin can simulate years of history to match your current standing crop.
- **Density:** The app suggests a standard density (plants/ha), but you can adjust this to match reality.

Step 3: 🦠 Disease Surveillance

Input observations made on the ground to "seed" the epidemiological model.

- **Select Threat:** Choose the primary disease concern (e.g., "Cocoa Black Pod", "Maize Streak Virus").
- **Mark Spots:** Click on the map to drop pins where infected plants were observed.
- **Set Severity:** In the table, log the number of infected plants at each spot.
    - *Why?* The model treats these as "Patient Zero" and uses wind/rain data to simulate how the pathogen spreads across your specific terrain over time.

Step 4: 🪨 Soil & Management

This step calibrates the engine to your specific soil conditions.

- **Satellite Auto-Detect:** Click 🛰️ **Auto-Detect Soil**. The app queries global geostatistical grids (OpenLandMap) to estimate:
    - **Texture:** (e.g., Sandy Clay Loam).
    - **Organic Carbon:** Used to calculate nutrient retention.
    - **Nitrogen Pools:** Estimates *Available N* based on a calculated C:N ratio.
- **Land Use History:** Use the slider to indicate how long the land has been farmed without fertilizer. This applies a "depletion factor" to the initial nutrient stocks.
- **Management Schedule:**
    - **Fertilizer:** Add events selecting specific commercial products (e.g., *"NPK 15-15-15 Compound"*, *"Urea"*).
    - **Operations:** For perennials, you can schedule events like **"Canopy Pruning"**.
    - **Irrigation:** Log any supplemental water added (in mm).

Step 5: 🚀 Launch

Review your summary and click 🔥 **Initialize Digital Twin**. This builds the simulation environment. Once complete, navigate to the **Intelligence Dashboard** tab.

🛰️ Intelligence Dashboard

This is your command center for monitoring the Digital Twin.

1. 📅 Timeline Controller

- **Slider:** Travel through time.
  - **Annual Crops:** View the season from planting to harvest.
  - **Perennial Crops:** View a **20-Year Horizon**, allowing you to see long-term biomass accumulation and annual cycles.
- **Past vs. Future:** Dates before "today" use historical weather data; dates after "today" use climate forecasts.

2. 🗺️ Spatial Epidemiology

Visualizes the field status for the selected date on a heatmap.

- **Disease Severity:** Red zones indicate high infection risk or hotspots.
- **Yield Potential:** Green zones show where biomass/fruit load is accumulating well.

3. 📈 Dynamics & Charts

Five tabs provide deep insight into crop physiology:

1. **Biomass & Growth:**
   - *Annuals:* Tracks Leaf Area Index (LAI) and Biomass.
   - *Perennials:* Shows the balance between **Structural Wood** (branches) and **Reproductive Fruit** (yield).
2. **Yield Forecast:** Shows the trajectory of yield accumulation. For perennials, you will see a "sawtooth" pattern representing annual harvests. The **Green Band** represents uncertainty due to climate variability.
3. **Nutrients & Stress:** Monitors the soil fuel tank (N, P, K levels) and stress spikes.
4. **Disease Pressure:** Tracks infection rates alongside "Environmental Favorability" (how friendly the weather is to the germ).
5. 🛰️ **Reality Check (NDVI):**
   - **Green Dashed Line:** What the model *predicts* the crop looks like.
   - **Blue Dots:** What the **Sentinel-2 Satellite** *actually* sees.
   - *Action:* If Blue Dots are far below the Green Line, inspect the field for unmodeled issues (pests, flooding, or animals).

📄 The Report

Navigate to the **Report** tab to generate a formal PDF dossier. *Note: Generating a report is computationally intensive because it runs multiple advanced engines:*

1. **Stochastic Ensemble:** Runs the simulation **50 times** with slightly different weather patterns to calculate confidence intervals (e.g., "Yield is 2.5 ± 0.3 tons").
2. **Irrigation Optimizer:** Automatically calculates the perfect watering schedule to eliminate the specific stress points found in your simulation.
3. **Nutrition Optimizer:** Analyses nutrient gaps and recommends a precise fertilizer schedule (Product + Rate + Date).

The PDF Dossier includes:

- **Executive Summary:** Total estimated production with uncertainty ranges.
- **Gap Analysis:** How much yield is being lost compared to the biophysical potential.
- **Smart Schedules:** A printable calendar for Irrigation (in Liters/ha) and Fertilization.
- **Risk Map:** A final map showing predicted disease severity at harvest.